

Section : Common

**Q.1** The corner points of the bounded feasible region determined by the system of linear inequalities are  $(0, 0)$ ,  $(4, 0)$ ,  $(2, 4)$  and  $(0, 5)$ . If maximum value of  $z = ax + by$ , where  $a, b > 0$ , occurs at both  $(2, 4)$  and  $(4, 0)$  then

1.  $a = 2b$
2.  $2a = b$
3.  $a = b$
4.  $3a = b$

Options 1. 1

2. 2
3. 3
4. 4

**Q.2** Area (in sq. units) of the region bounded by the curve  $y^2 = 4x$ , y-axis and the line  $y = 3$  is

1.  $\frac{9}{4}$
2. 2
3. 3
4.  $\frac{9}{2}$

Options 1. 1

2. 2
3. 3
4. 4

**Q.3** The function  $f(x) = -\frac{3}{4}x^4 - 8x^3 - \frac{45}{2}x^2 + 163$  has a local maxima at

1.  $x = 0$  and  $x = 3$
2.  $x = -3$  and  $x = -5$
3.  $x = 0$  and  $x = -5$
4.  $x = -3$

Options 1. 1

2. 2
3. 3
4. 4

**Q.4** If a matrix has 8 elements then the possible order(s) it may have

- (A)  $8 \times 1$
- (B)  $5 \times 3$
- (C)  $6 \times 2$
- (D)  $2 \times 4$

Choose the \textbf{correct} answer from the options given below:

1. (A) and (D) only
2. (B) and (C) only
3. (A) only
4. (A), (B), (C) and (D)

Options 1. 1

2. 2
3. 3
4. 4

**Q.5** The minimum value of  $Z = 2x + y$  subjected to  $x + y \geq 10$ ,  $2x + 3y \leq 26$ ,  $x, y \geq 0$  is

1. 14
2. 20
3. 26
4. 30

Options 1. 1

2. 2
3. 3
4. 4

**Q.6** The solution of the differential equation  $\frac{dr}{dt} = -rt$ ,  $r(0) = r_0$  is

1.  $r = r_0 e^{t^2/2}$
2.  $r = -r_0 e^{t^2/2}$

3.  $r = r_0 e^{-t^2/2}$

4.  $r = -r_0 e^{-t^2/2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.7

Let  $x = t^2, y = t^3$ . Then  $\frac{d^2y}{dx^2}$  is equal to

1.  $\frac{3}{2}$

2.  $\frac{3}{4t}$

3.  $3t$

4.  $\frac{3t}{4}$

Options 1. 1

2. 2

3. 3

4. 4

Q.8

$\int_0^1 x e^x dx$  is equal to

1. 0

2. 1

3.  $e$

4.  $e^2$

Options 1. 1

2. 2

3. 3

4. 4

Q.9

The function  $f(x) = x^2 e^{-2x}$  increases on

1.  $(-\infty, \infty)$

2.  $(0, 1)$

3.  $(-\infty, 0) \cup (1, \infty)$

4.  $(1, \infty)$

Options 1. 1

2. 2

3. 3

4. 4

Q.10

$\int \frac{x}{(x-1)(x-2)} dx$  is equal to ( where C is a constant of integration)

1.  $\log_e \left| \frac{(x-1)^2}{x-2} \right| + C$

2.  $\log_e \left| \frac{(x-2)^2}{x-1} \right| + C$

3.  $\log_e |(x-2)(x-1)^2| + C$

4.  $\log_e \left| \frac{x-1}{(x-2)^2} \right| + C$

Options 1. 1

2. 2

3. 3

4. 4

Q.11

Let A be a non-singular square matrix of order 3 and  $|\text{adj } A| = 5$  then  $|A|$  is equal to

1.  $\pm \sqrt{5}$

2. 5

3. 25

4. 125

Options 1. 1

2. 2

3. 3

4. 4

Q.12

If  $A = \begin{bmatrix} a & a & a \\ 0 & a & a \\ 0 & 0 & a \end{bmatrix}$ , then  $|\text{adj } A|$  is equal to

1.  $a^3$
2.  $a^6$
3.  $a^9$
4.  $a^{27}$

Options 1. 1

2. 2
3. 3
4. 4

Q.13

If  $A$  is a square matrix and  $I$  is an identity matrix of same order such that  $A^2 = A$ , then  $(I + A)^3 - 8I$  is equal to

1.  $A$
2.  $A - I$
3.  $7(A - I)$
4.  $I$

Options 1. 1

2. 2
3. 3
4. 4

Q.14

Let  $X$  denotes the number of doublets obtained in 3 throws of a pair of dice.

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \quad \textbf{List-II}

(A)  $P(X = 0)$  (I)  $\frac{1}{216}$

(B)  $P(X = 1)$  (II)  $\frac{15}{216}$

(C)  $P(X = 2)$  (III)  $\frac{75}{216}$

(D)  $P(X = 3)$  (IV)  $\frac{125}{216}$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (II), (C) - (III), (D) - (I)
3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
4. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.15

Match \textbf{List-I} with \textbf{List-II}

**List-I**

**List-II**

**Differential Equation**

**Degree**

(A)  $xy \frac{d^2y}{dx^2} + x \left( \frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$

(I) 3

(B)  $\frac{d^2y}{dx^2} + \log \left( \frac{dy}{dx} \right) = 0$

(II) 1

(C)  $\left( \frac{d^2y}{dx^2} \right)^2 + \left( \frac{dy}{dx} \right)^3 + \frac{dy}{dx} + 1 = 0$

(III) not defined

(D)  $2x^2 \left( \frac{d^2y}{dx^2} \right)^3 - 5 \left( \frac{dy}{dx} \right)^2 + y = 0$

(IV) 2

Choose the \textbf{correct} answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

3. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Section : Applied Mathematics

Q.16 Which one of the following statement is not correct?

1. Sample is a collection of some elements of population having same characteristic.

2. The measurable characteristic of sample is called statistic.

3. A survey done using a sample of the population has no margin of error.

4. The sample makes inference about the population.

Options 1. 1

2. 2

3. 3

4. 4

Q.17 Five dice are thrown simultaneously. If the occurrence of an even number in a single dice is considered a success, then the probability of at most 3 successes is:

1.  $\frac{15}{16}$

2.  $\frac{13}{16}$

3.  $\frac{11}{16}$

4.  $\frac{1}{16}$

Options 1. 1

2. 2

3. 3

4. 4

Q.18 The equation of the tangent line to the curve  $y = x^2 - 2x + 5$  which is parallel to the line  $4x - y + 1 = 0$  is:

1.  $4x - y - 1 = 0$

2.  $4x - y - 4 = 0$

3.  $x + 4y - 4 = 0$

4.  $x + 4y - 1 = 0$

Options 1. 1

2. 2

3. 3

4. 4

Q.19 The differential equation of the family of curves  $y = Ae^{3x} + Be^{-3x}$ , where  $A$  and  $B$  are arbitrary constants, is

1.  $\frac{d^2y}{dx^2} + 9y = 0$

2.  $\frac{d^2y}{dx^2} + 3y = 0$

3.  $\frac{d^2y}{dx^2} = 9y$

4.  $\frac{d^2y}{dx^2} = 3y$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.20

The unit's digit in  $13^{61}$  is

- 1. 1
- 2. 3
- 3. 7
- 4. 9

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.21

If  $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} m & 14 \\ 2 & n \end{bmatrix}$ , then  $m + n$  is equal to:

- 1. 0
- 2. 22
- 3. 16
- 4. 36

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.22 Kavya takes a personal loan of ₹ 10,00,000 at the rate of 12% per annum for

5 years, then the EMI by using flat rate method is:

- 1. ₹ 26,666.66
- 2. ₹ 36,666.66
- 3. ₹ 16,666.66
- 4. ₹ 46,666.66

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.23 The solution set of the inequation  $4x + 3y > 5$  is

- 1. half plane that contains the origin
- 2. open half plane not containing the origin
- 3. whole  $xy$ -plane except the points lying on the line  $4x + 3y = 5$
- 4. lying only on first quadrant

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.24

If  $\frac{1}{x^2} - \frac{1}{x} > 0$ , then  $x$  lies in the interval

- 1.  $(-\infty, 0)$
- 2.  $(-\infty, 0) \cup (0, 1)$
- 3.  $(-\infty, 1)$
- 4.  $(1, \infty)$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.25 If  $X$  is a random variable with probability distribution as given below:

X 0 1 2 3

P(X)  $\frac{1}{4}$   $\frac{1}{2}$   $\frac{1}{4}$   $\frac{1}{4}$

Then, the variance of the distribution is:

1.  $\frac{65}{51}$

2.  $\frac{72}{49}$

3.  $\frac{62}{49}$

4.  $\frac{62}{51}$

Options 1. 1

2. 2

3. 3

4. 4

Q.26 The following data are from a simple random sample: 6, 8, 11, 9, 15, 17,

then the point estimate of the population mean is:

1. 9

2. 10

3. 11

4. 12

Options 1. 1

2. 2

3. 3

4. 4

Q.27 Consider the following hypothesis test:

$$H_0 : \mu \leq 16$$

$$H_a : \mu > 16$$

A sample of 25 provided a sample mean  $\bar{x} = 17$  and a sample standard deviation  $S = 3.5$ , then the value of the test statistic in  $t$ -test is:

1. 1.03

2. 1.33

3. 1.43

4. 1.53

Options 1. 1

2. 2

3. 3

4. 4

Q.28 The sale of ice creams is higher in summer than in winter is an example of:

1. Seasonal trend

2. Cyclical trend

3. Long term trend

4. Irregular trend

Options 1. 1

2. 2

3. 3

4. 4

Q.29 The number of all possible matrices of order  $2 \times 2$  with each entry 0, 1 or 2 are:

1. 64

2. 81

3. 8

4. 9

Options 1. 1

2. 2

3. 3

4. 4

Q.30 The derivative of  $(\log x)^x$  with respect to  $\log x$  is:

1.  $x(\log x)^x [1 + \log x \cdot \log(\log x)]$

2.  $x(\log x)^x [x + 1 + \log x \cdot \log(\log x)]$

3.  $x(\log x)^{x-1} [1 + \log x \cdot \log(\log x)]$

4.  $(\log x)^{x-1} [1 + \log x \cdot \log(\log x)]$

Options 1. 1

2. 2

3. 3

4. 4

Q.31 Two runners A and B complete a 150 meters race in 24 seconds and 36 seconds respectively.

By how many meters will A defeat B?

1. 100 meters

2. 20 meters

3. 25 meters

4. 50 meters

Options 1. 1

2. 2

3. 3

4. 4

Q.32 If  $A = \begin{bmatrix} 7 & 3 \\ 5 & -7 \end{bmatrix}$  be such that  $A^{-1} = kA$ , then  $k$  equals

1.  $-\frac{1}{64}$

2.  $\frac{1}{64}$

3. 64

4.  $-64$

Options 1. 1

2. 2

3. 3

4. 4

Q.33

Match List-I with List-II

| List-I                     | List-II                    |
|----------------------------|----------------------------|
| Function                   | Increasing on the interval |
| (A) $f(x) = -x^2 - 2x + 1$ | (I) $(-\infty, -1)$        |
| (B) $f(x) = x^2 + 1$       | (II) $(1, \infty)$         |
| (C) $f(x) = x^2 - 2x + 3$  | (III) $(-\infty, 0)$       |
| (D) $f(x) = -x^2$          | (IV) $(0, \infty)$         |

Choose the **correct** answer from the options given below:

- (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.34 A man rows 24 km upstream in 6 hours and 36 km downstream in 6 hours each time, then the speed of the boat in the still water is

- 5 km/hr
- 1 km/hr
- 6 km/hr
- 4 km/hr

Options 1. 1

2. 2

3. 3

4. 4

Q.35 Which one of the following statement is **incorrect** about sinking fund?

- It is a fixed term account.
- It is set-up for a particular upcoming expense.
- In this fund a fixed amount at regular intervals is deposited.
- It can be used in any emergency.

Options 1. 1

2. 2

3. 3

4. 4

Q.36 For the given five values 16, 25, 19, 34, 46, the three years moving averages are:

- 20, 26, 33
- 20, 26, 32
- 21, 25, 33

4. 20, 25, 34

Options 1. 1

2. 2

3. 3

4. 4

Q.37 For what value of  $k$ , the following system have a unique solution? (where  $\mathbb{R}$  is set of real numbers)

$$x + y + z = 1$$

$$2x + 3y + 4z = 3$$

$$x - y + kz = 5$$

1.  $k \in \mathbb{R} \sim \{3\}$

2.  $k \in \mathbb{R} \sim \{-3\}$

3.  $k \in \mathbb{R} \sim \{4\}$

4.  $k \in \mathbb{R} \sim \{-4\}$

Options 1. 1

2. 2

3. 3

4. 4

Q.38 A random variable  $X$  has the following probability distribution:

$X$     -2   -1   0   1   2   3

$P(X)$  0.1 0.2 0.3 0.2 0.1 0.1

Choose the correct answer from the options given below:

Unknown environment 'enumerate'

Options 1. 1

2. 2

3. 3

4. 4

Q.39 The demand function for a certain product is represented by the equation:  $p = 20 + 5x - 3x^2$ ,

where  $x$  is the number of units demanded and  $p$  is the price per unit (in Rs.),

then the marginal revenue when 2 units are sold, is:

1. Rs. 8

2. Rs. 4

3. Rs. 6

4. Rs. 2

Options 1. 1

2. 2

3. 3

4. 4

Q.40 If  $n$  is the sample size of the population, then degree of freedom in t-distribution is

1.  $n$

2.  $2n$

3.  $n + 1$

4.  $n - 1$

Options 1. 1

2. 2

3. 3

4. 4

**Q.41** If  $A$  and  $B$  are symmetric matrices of the same order, then which one of the following is true?

1.  $A + B$  is a skew-symmetric matrix

2.  $A - B$  is skew-symmetric matrix.

3.  $AB - BA$  is a symmetric matrix.

4.  $AB + BA$  is a symmetric matrix.

Options 1. 1

2. 2

3. 3

4. 4

**Q.42** The minimum value of  $Z = 3x + 2y$  subjected to the constraints  $2x + y \geq 7$ ,  $x + 2y \geq 8$ ,  $x, y \geq 0$  is:

1. 24

2. 10

3. 12

4. 14

Options 1. 1

2. 2

3. 3

4. 4

**Q.43** In what ratio must pulses at ₹72 per kg be mixed with pulses ₹90 per kg so that the mixture be worth ₹80 per kg?

1. 5:6

2. 5:4

3. 4:5

4. 5:8

Options 1. 1

2. 2

3. 3

4. 4

**Q.44** The term of the perpetuity is:

1. Infinite

2. finite

3. between 0 to 100

4. we can't say about term.

Options 1. 1

2. 2

3. 3

4. 4

**Q.45** The integral  $\int e^x \left( \frac{x-1}{2x^2} \right) dx$  is equal to

1.  $\frac{e^x}{2x} + C$ , where  $C$  is constant of integration.

2.  $\frac{2e^x}{x} + C$ , where  $C$  is constant of integration.

3.  $\frac{e^x}{x} + C$ , where  $C$  is constant of integration.

4.  $\frac{e^x}{4x} + C$ , where  $C$  is constant of integration.

Options

1. 1
2. 2
3. 3
4. 4

**Q.46** Pipe A can fill a tank in 24 hours and pipe B in 32 hours. If both the pipes are opened in an empty tank at the same time, then the time taken to fill the tank is:

1.  $13\frac{3}{7}$  hours

2.  $13\frac{2}{7}$  hours

3.  $13\frac{5}{7}$  hours

4.  $13\frac{1}{7}$  hours

Options

1. 1
2. 2
3. 3
4. 4

**Q.47** The mean of the number of heads in the two tosses of a coin is

1. 1

2. 2

3. 4

4. 5

Options

1. 1
2. 2
3. 3
4. 4

**Q.48** The Value of  $\int_1^3 |2x - 1| dx$  equal to

1. 2

2. 4

3. 6

4. 8

Options

1. 1
2. 2
3. 3
4. 4

Q.49

Consider the following data:

| Year (x)       | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 |
|----------------|------|------|------|------|------|------|------|
| Profit (₹) (y) | 114  | 130  | 126  | 144  | 138  | 156  | 164  |

The equation of straight line trend by using least square method is,  $y = 138.86 + 7.64(x - 2007)$ , then the trend values for the year 2004 is

1. 115.94
2. 215.94
3. 151.94
4. 111.94

Options 1. 1

2. 2
3. 3
4. 4

Q.50 A machine costing ₹1,00,000 has a useful life of 5 years. The estimated scrap value is ₹20,000.

Using straight line method, the annual depreciation is:

1. ₹12,000
2. ₹16,000
3. ₹20,000
4. ₹10,000

Options 1. 1

2. 2
3. 3
4. 4

#### Section : Core Mathematics

Q.51 If A is a skew-symmetric matrix, then which of the following statements is **NOT** true?

- (A) A is singular if order of A is odd
- (B) A is non-singular
- (C)  $A^{2025}$  is a skew-symmetric matrix
- (D)  $A^{2025}$  is a symmetric matrix
- (E) all diagonal elements of A are zeros

Choose the **correct** answer from the options given below:

1. (A), (C) and (E) only
2. (A) and (E) only
3. (B), (D) and (E) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.52 Let  $[t]$  denote the greatest integer  $\leq t$  and  $a\mathbb{Z} = \{ax : x \in \mathbb{Z}, a \in \mathbb{R}\}$  (where  $\mathbb{Z}$  is set of integer and  $\mathbb{R}$  is set of real number). The set of points of discontinuity of the function  $f(x) = [2x]$  is given by

1.  $\mathbb{Z}$
2.  $2\mathbb{Z}$
3.  $\frac{1}{2}\mathbb{Z}$
4.  $\mathbb{R} - \mathbb{Z}$

Options 1. 1

2. 2
3. 3
4. 4

Q.53 The area (in sq. units) of the region bounded by the curve  $y = \sqrt{16 - x^2}$  and x-axis is

1.  $8\pi$
2.  $16\pi$

3.  $24\pi$

4.  $32\pi$

Options 1. 1

2. 2

3. 3

4. 4

Q.54

If  $A = \begin{bmatrix} 0 & 1 & -3 \\ -1 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}$ , then the value of  $|A^{2025}|$  is

1. 1

2. 126

3.  $-126$

4. 0

Options 1. 1

2. 2

3. 3

4. 4

Q.55

The area (in sq. units) bounded by the curve  $y = \cos x$  and  $x$ -axis between  $x = 0$  and  $x = \frac{3\pi}{2}$  is

1. 1

2. 2

3. 3

4. 4

Options 1. 1

2. 2

3. 3

4. 4

Q.56

Which of the following statements is/are true?

(A)  $(\tan^{-1} y - x)dy = (1 + y^2)dx$  is a differential equation where variables are separable.

(B)  $(1 + x^2)dy + 2xydx = \cot x dx (x \neq 0)$  is a first order linear differential equation.

(C)  $(4x + 6y + 5)dy - (3y + 2x + 4)dx = 0$  is not a homogeneous differential equation.

(D)  $(xy)dx - (x + y^2)dy = 0$  is a homogeneous differential equation.

Choose the \textbf{correct} answer from the options given below:

1. (A) only

2. (B) and (C) only

3. (B) and (D) only

4. (A), (B), (C) and (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.57

If E and F are independent events associated with an experiment, then which one of the following statements is correct?

1.  $P(E \cap F) = 0$

2.  $P(E \cup F) = 1$

3.  $P(E|F) = P(E)$

4.  $P(E|F) = P(F)$

Options 1. 1

2. 2

3. 3

4. 4

Q.58

A and B are two sets such that  $n(A) = 5$  and  $n(B) = 7$ . The number of one-one functions from A to B is

1. 35

2. 2150

3. 2520

4. 120

Options 1. 1

2. 2

3. 3

4. 4

Q.59 If  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  be vectors such that  $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ ,  $|\vec{a}| = 3$ ,  $|\vec{b}| = 5$  and  $|\vec{c}| = 7$ , then the angle between  $\vec{a}$  and  $\vec{b}$  is

1.  $\frac{\pi}{2}$
2.  $\frac{\pi}{3}$
3.  $\frac{\pi}{4}$
4.  $\frac{\pi}{6}$

Options 1. 1

2. 2
3. 3
4. 4

Q.60  $\int \frac{dx}{2\sin^2 x + 5\cos^2 x}$  is equal to

1.  $\frac{1}{\sqrt{10}} \tan^{-1} \left( \frac{\tan x}{\sqrt{5}} \right) + C$ ; C is an arbitrary constant
2.  $\frac{1}{\sqrt{5}} \tan^{-1} \left( \frac{2\tan x}{\sqrt{5}} \right) + C$ ; C is an arbitrary constant
3.  $\frac{1}{\sqrt{2}} \tan^{-1} \left( \frac{\sqrt{2}\tan x}{\sqrt{5}} \right) + C$ ; C is an arbitrary constant
4.  $\frac{1}{\sqrt{10}} \tan^{-1} \left( \frac{\sqrt{2}\tan x}{\sqrt{5}} \right) + C$ ; C is an arbitrary constant

Options 1. 1

2. 2
3. 3
4. 4

Q.61 If a line makes angle  $\pi/3$  and  $\pi/4$  with the positive directions of  $x$ -axis and  $y$ -axis respectively, then the acute angle made by the line with positive direction of  $z$ -axis is

1.  $\pi/2$
2.  $\pi/4$
3.  $\pi/3$
4.  $2\pi/3$

Options 1. 1

2. 2
3. 3
4. 4

Q.62 Which of the following statements is/are true?

(A) The vector sum of the three sides of a triangle in order is  $\vec{0}$

(B) The magnitude ( $r$ ), direction ratios ( $a, b, c$ ) and direction cosines ( $l, m, n$ ) of any vector  $\vec{r} = a\hat{i} + b\hat{j} + c\hat{k}$  are related as  $l = \frac{a}{r}, m = \frac{b}{r}, n = \frac{c}{r}$

(C) If  $\theta$  is the angle between two vectors  $\vec{a}$  and  $\vec{b}$ , then their cross product is given as  $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta$

(D) The cross product of two vectors is commutative

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A) and (B) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.63 If  $f(a-x) = f(x)$ , then  $\int_0^a xf(x)dx$  is equal to

1.  $\frac{a}{2} \int_0^a f(x)dx$
2.  $2 \int_0^a f(x)dx$
3.  $af(a)$
4.  $a \int_0^a f(x)dx$

Options 1. 1

2. 2

3. 3

4. 4

Q.64 Consider the LPP: Max  $Z = 5x + 3y$  subject to  $3x + 5y \leq 15$ ,  $5x + 2y \leq 10$ ,  $x \geq 0$ ,  $y \geq 0$

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I}

\textbf{List-II}

(A) Objective function

(I)  $3x + 5y \geq 15$

(B) One constraint

(II)  $x, y \geq 0$

(C) Non-negative restrictions

(III)  $Z = 5x + 3y$

(D) Point (1, 2) does not lie in the region (IV)  $3x + 5y \leq 15$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

3. (A) - (III), (B) - (I), (C) - (II), (D) - (IV)

4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.65 The number of equivalence relation on the set  $\{1, 2, 3\}$  containing (1, 2) and (2, 1) is

1. 1

2. 2

3. 3

4. 4

Options 1. 1

2. 2

3. 3

4. 4

Q.66 Probability that a man speaks truth is  $\frac{3}{4}$ . He throws a die and reports that it is a six. The probability that it is actually a

six is

1.  $\frac{1}{3}$

2.  $\frac{3}{8}$

3.  $\frac{5}{8}$

4.  $\frac{2}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.67 If  $y = \sin^{-1} x + \sin^{-1} \sqrt{1 - x^2}$ ,  $x \in (-1, 0)$ , then  $\frac{dy}{dx}$  is equal to

1. 0

2.  $\frac{1}{\sqrt{1 - x^2}}$

3.  $\frac{2}{\sqrt{1 - x^2}}$

4.  $\frac{-2}{\sqrt{1 - x^2}}$

Options 1. 1

2. 2

3. 3

4. 4

Q.68

The value(s) of  $K$ , for which the system of linear equations  $2x + y + z = 1$ ,  $x + Ky - z = \frac{3}{2}$  and  $3y - 5z = 9$  does not possess a unique solution is

1.  $K = \frac{7}{5}$
2.  $K = -\frac{7}{5}$
3.  $K \in R - \left\{ \frac{7}{5} \right\}$
4.  $K \in R - \left\{ -\frac{7}{5} \right\}$

Options 1. 1

2. 2
3. 3
4. 4

Q.69

The minimum value of the function  $f(x) = 3 \sin x - 4 \cos x$ ,  $x \in [-4\pi, 4\pi]$  is equal to

1. 0
2. -4
3. -5
4. -7

Options 1. 1

2. 2
3. 3
4. 4

Q.70

Consider two independent events A and B such that  $P(A) = 0.3$ ,  $P(B) = 0.6$ .

Match \textbf{List-I} with \textbf{List-II}

**List-I****List-II**

- |                                           |            |
|-------------------------------------------|------------|
| (A) $P(A \text{ and } B)$                 | (I) 0.28   |
| (B) $P(A \text{ and not } B)$             | (II) 0.18  |
| (C) $P(A \text{ or } B)$                  | (III) 0.12 |
| (D) $P(\text{neither } A \text{ nor } B)$ | (IV) 0.72  |

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
2. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
3. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.71

$\int \left( \frac{1}{\log_e x} - \frac{1}{(\log_e x)^2} \right) dx$  is equal to

1.  $\frac{x}{\log_e x} + c$  :  $C$  is an arbitrary constant
2.  $\frac{1}{\log_e x} + c$  :  $C$  is an arbitrary constant
3.  $\frac{x}{(\log_e x)^2} + c$  :  $C$  is an arbitrary constant
4.  $\frac{\log_e x}{x} + c$  :  $C$  is an arbitrary constant

Options 1. 1

2. 2
3. 3
4. 4

Q.72

The rate of change of volume of a sphere with respect to its surface area, when radius is 4 cm, is equal to

1.  $64\pi \text{ cm}^3/\text{cm}^2$
2.  $32\pi \text{ cm}^3/\text{cm}^2$
3.  $\frac{1}{2} \text{ cm}^3/\text{cm}^2$

4.  $2 \text{ cm}^3/\text{cm}^2$

Options 1. 1

2. 2

3. 3

4. 4

Q.73 The corner points of the feasible region determined by a system of linear constraints are  $(0, 0)$ ,  $(0, 40)$ ,  $(20, 40)$ ,  $(60, 20)$ ,  $(60, 0)$ . If the objective function is  $Z = 4x + 3y$ , then which one of the following is true?

1.  $\text{Max } Z > 400$

2.  $\text{Max } Z < 400$

3.  $\text{Max } Z = 400$

4.  $\text{Max } Z = 350$

Options 1. 1

2. 2

3. 3

4. 4

Q.74 Which of the following functions  $f(x)$  are differentiable at  $x = 0$ ?

(A)  $|x|$

(B)  $|x - 1|$

(C)  $[x]$ , where  $[t]$  denotes the greatest integer  $\leq t$

(D)  $|x + 1|$

(E)  $x^2$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (D) only

2. (B) and (D) only

3. (B), (D) and (E) only

4. (A), (D) and (E) only

Options 1. 1

2. 2

3. 3

4. 4

Q.75 Let  $A$  be a square matrix of order  $n$ , then which of the following are TRUE?

(A)  $|\text{adj } A| = |A|^{n-1}$

(B)  $|A \cdot \text{adj } A| = |A|^n$

(C)  $A \cdot (\text{adj } A) = |A|$

(D)  $|KA| = K|A|$

(E)  $|A^{-1}| = \frac{1}{|A|}, |A| \neq 0$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (E) only

2. (A), (B), (C) and (E) only

3. (B), (C) and (D) only

4. (C), (D) and (E) only

Options 1. 1

2. 2

3. 3

4. 4

Q.76 The value of  $\tan^2(\sec^{-1} 2) + \cot^2(\text{cosec}^{-1} 3)$  is equal to

1. 6

2. 11

3. 13

4. 15

Options 1. 1

2. 2

3. 3

4. 4

Q.77 Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I}

\textbf{List-II}

(A) Equations of line through (5, -4, 6) with direction ratios 3, 7, 2

$$(I) \frac{x+3}{5} = \frac{y+7}{-4} = \frac{z+2}{6}$$

(B) Equations of line through (3, 7, 2) with direction ratios 5, -4, 6

$$(II) \frac{x-3}{5} = \frac{y-7}{-4} = \frac{z-2}{6}$$

(C) Equations of line through (-5, 4, -6) with direction ratios 3, 7, 2

$$(III) \frac{x-5}{3} = \frac{y+4}{7} = \frac{z-6}{2}$$

(D) Equations of line through (-3, -7, -2) with direction ratios 5, -4, 6

$$(IV) \frac{x+5}{3} = \frac{y-4}{7} = \frac{z+6}{2}$$

Choose the correct answer from the options given below:

- (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
- (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
- (A) - (IV), (B) - (I), (C) - (III), (D) - (II)
- (A) - (I), (B) - (IV), (C) - (II), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.78

Let  $A = \begin{bmatrix} 152 & 105 & 3 \\ 149 & 25 & 35 \\ 2 & 1 & 0 \end{bmatrix}$ . If  $A_{ij}$  denotes the co-factor of an element  $a_{ij}$  of the matrix  $A$ , then the value of

$a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23}$  is equal to

1. -1
2. 0
3. 1
4. 1529

Options 1. 1

2. 2
3. 3
4. 4

Q.79

Two persons A and B throw a die alternately till one of them gets a six and wins the game. If A begins, then the probabilities of winning of A and B respectively are

1.  $\frac{1}{6}, \frac{1}{6}$
2.  $\frac{2}{3}, \frac{1}{3}$
3.  $\frac{6}{11}, \frac{5}{11}$
4.  $\frac{5}{11}, \frac{6}{11}$

Options 1. 1

2. 2
3. 3
4. 4

Q.80

If  $\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ x \end{bmatrix} = 0$ , then value of  $x$  is

1. -1
2. 0
3. 1
4. 2

Options 1. 1

2. 2
3. 3
4. 4

Q.81

If the lines  $\frac{1-x}{3} = \frac{y-2}{2\lambda} = \frac{z-3}{2}$  and  $\frac{x-1}{3\lambda} = \frac{y-1}{1} = \frac{6-z}{5}$  are perpendicular, then  $\lambda$  is equal to

1.  $\frac{10}{11}$
2.  $\frac{-10}{7}$
3.  $\frac{-10}{11}$
4.  $\frac{10}{7}$

4.  $\frac{10}{7}$

Options 1. 1

2. 2

3. 3

4. 4

Q.82 The function  $f(x) = \log_e(\sin x)$ ,  $x \in (0, \pi)$  is

(A) strictly increasing on  $(0, \frac{\pi}{2})$

(B) strictly decreasing on  $(0, \frac{\pi}{2})$

(C) strictly increasing on  $(\frac{\pi}{2}, \pi)$

(D) strictly decreasing on  $(\frac{\pi}{2}, \pi)$

(E) strictly increasing on  $(0, \pi)$

Choose the \textbf{correct} answer from the options given below:

1. (A) and (D) only

2. (B) and (C) only

3. (A), (D) and (E) only

4. (B), (D) and (E) only

Options 1. 1

2. 2

3. 3

4. 4

Q.83 Let  $\vec{a} = \hat{i} + \hat{j} + \hat{k}$  and  $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$ , then a unit vector perpendicular to both vectors  $(\vec{a} + \vec{b})$  and  $(\vec{a} - \vec{b})$  is equal to

1.  $\frac{1}{\sqrt{6}}(-\hat{i} + 2\hat{j} - \hat{k})$

2.  $\frac{1}{\sqrt{6}}(\hat{i} - 2\hat{j} - \hat{k})$

3.  $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

4.  $\frac{1}{\sqrt{6}}(\hat{i} + 2\hat{j} + \hat{k})$

Options 1. 1

2. 2

3. 3

4. 4

Q.84 Let  $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$  and  $\vec{b} = -2\hat{i} + 3\hat{j} - 4\hat{k}$ , then which of the following statements are correct ?

(A)  $|\vec{a}| = \sqrt{14}$

(B)  $|\vec{b}| = 29$

(C)  $\vec{a} \cdot \vec{b} = 8$

(D) Angle between  $\vec{a}$  and  $\vec{b}$  is  $\cos^{-1}\left(\frac{-8}{\sqrt{406}}\right)$

Choose the \textbf{correct} answer from the options given below:

1. (A) and (D) only

2. (A) and (C) only

3. (B), (C) and (D) only

4. (A), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.85 The solution of the differential equation  $\frac{dy}{dx} - \frac{y}{x} = 2 \log_e x$

1.  $y = x \log_e x + C$  :  $C$  is an arbitrary constant

2.  $y = x(\log_e x + C)$  :  $C$  is an arbitrary constant

3.  $y = x((\log_e x)^2 + C)$  :  $C$  is an arbitrary constant

4.  $y = x(2(\log_e x)^2 + C)$  :  $C$  is an arbitrary constant

Options 1. 1

2. 2

3. 3

